Neurosymbolic Knowledge Engineering with Natural Language

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PhD Defence Layman's Talk

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The implementation problem in knowledge engineering

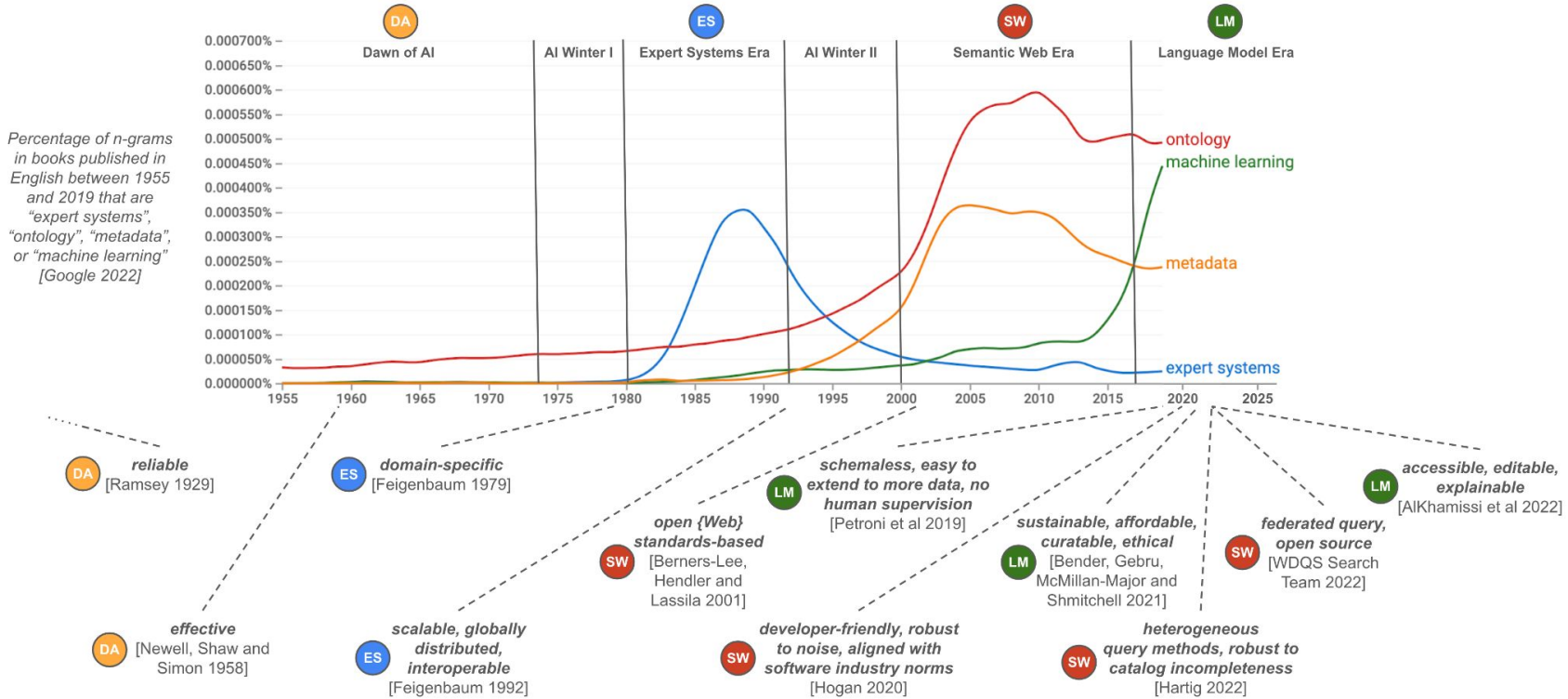


Edward Feigenbaum

“The expert is manually representing knowledge for the knowledge base; the user is exploiting the interaction to get problem solutions. ***I told you that I started some AI companies – the reason I am not wealthy right now – the reason that I don’t have tens of millions of dollars… is that bottleneck – it was too hard to get knowledge into the knowledge bases.”***

M.-S. Augier and M. T. Vendelø, “An Interview with Edward A. Feigenbaum,” Copenhagen Business School, Department of Informatics, Copenhagen, Denmark, Interview, 2002.

The problem has persisted for over 50 years



Could LLMs provide a solution?



Emily Bender

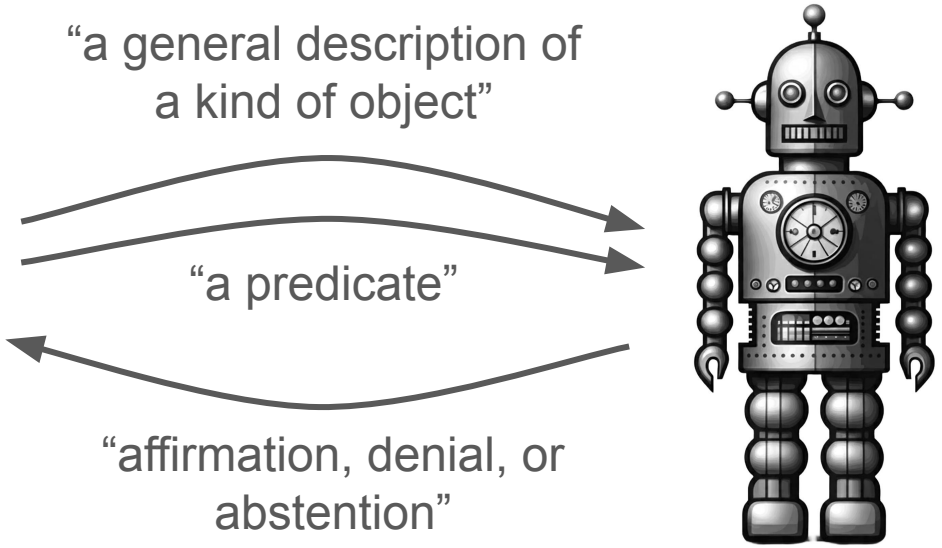
Could domain experts use LLMs to build knowledge bases with natural language, directly addressing the implementation problem?

But wait: aren't LLMs just *stochastic parrots*, whose output is without meaning, and riddled with inconsistency and gaps in knowledge?

Inspiration: Carnap's Robot



Rudolf Carnap



R. Carnap, "Meaning and synonymy in natural languages," Philosophical studies, vol. 6, pp. 33–47, 1955.

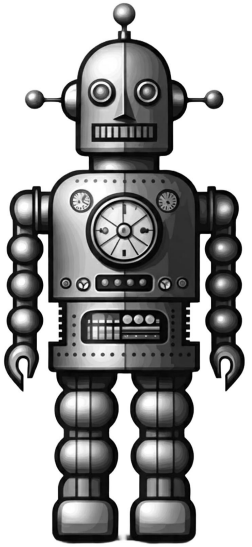
A neurosymbolic approach: LLM as Carnap's Robot



system prompt + context

atomic statement

truth value with rationale



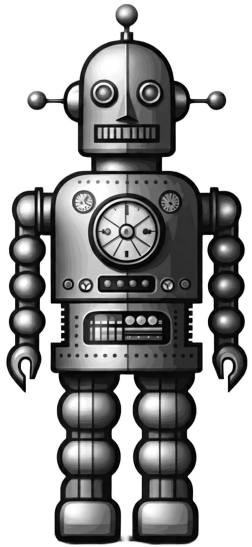
Using natural language to classify instances

intensional concept definition

A planet [1] is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, and (c) has cleared the neighbourhood around its orbit.

instance description

2MASS J03552337+1133437 (2MASS J0355+11) is a nearby brown dwarf of spectral type L5γ, located in constellation Taurus at approximately 29.8 light-years from Earth.



classification with rationale

The description of 2MASS J03552337+1133437 identifies it as a brown dwarf, not a planet. Furthermore, the definition requires a planet to be in orbit around the Sun, and there is no information provided that 2MASS J03552337+1133437 orbits the Sun. **Therefore, based on the provided definition and information, we cannot argue that 2MASS J03552337+1133437 is a planet.**

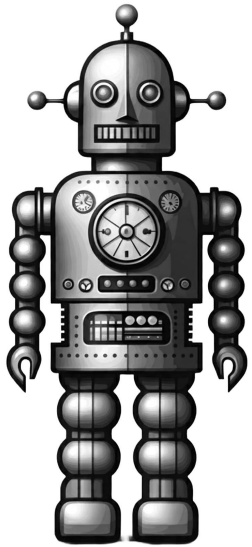
Using natural language to validate knowledge graphs

intensional property definition

made from material: material the subject or the object is made of or derived from (do not confuse with P10672 which is used for processes)

knowledge graph triple

Subject: chocolate (Q195)
Predicate: made from material (P186)
Object: sugar (Q11002)



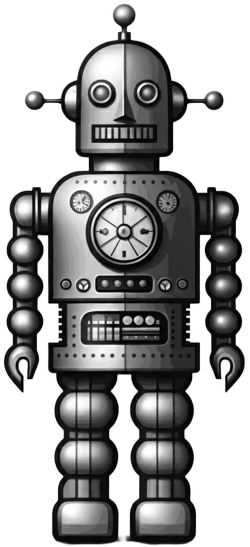
classification with rationale

There is no mention of sugar being a material that chocolate is made from. In fact, the passage states that sugar is added to chocolate to create sweet chocolate, but it is not a primary ingredient that chocolate is made from. **Therefore, the given knowledge graph triple is false**.

Flagging gluts and gaps in LLM knowledge

atomic statement

America was discovered in 1492



verifiability classification

The answer is contextually accurate within the framework of European exploration history, but it does not encompass the full scope of human discovery of the Americas. **CANNOT VERIFY**

refutability classification

While it does not account for earlier discoveries by indigenous peoples or Norse explorers, it is not a contradiction to state that America was discovered in 1492 from a European historical perspective. **CANNOT REFUTE**

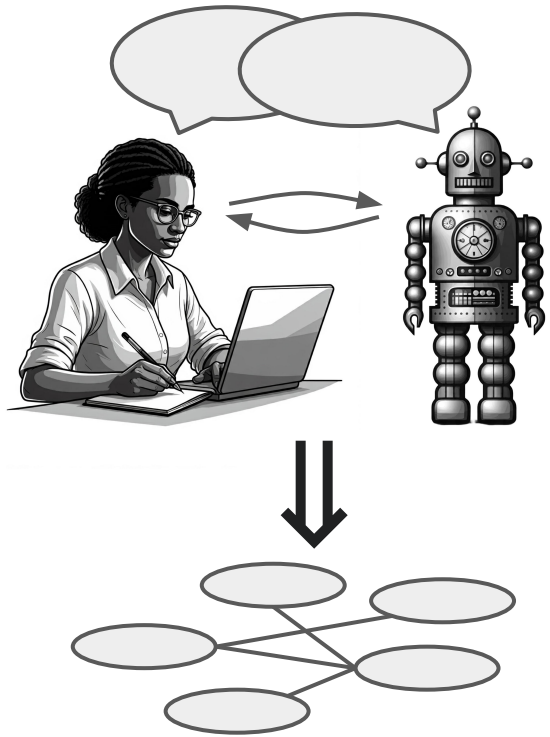
bilateral truth value

$\langle \tilde{f}, \hat{f} \rangle$ (*gap*)

Our key findings

Research question	Finding	Empirical results
Can LLM-interpreted natural language definitions represent concepts?	**Yes**, with near-perfect agreement — LLM judgments often more accurate than the knowledge graph	● Cohen's κ: 0.92–0.96 (N=200) ● 67% of misclassifications were knowledge graph errors
Can they be used to validate and refine relations in knowledge graphs?	**Yes**, with high precision and ability to identify knowledge graph errors — both factual and metalinguistic	● F1: 0.830 (Wikidata), 0.893 (CaLiGraph) (N=1560) ● 40.9% of disagreements due to knowledge graph errors ● Mean metalinguistic disagreement rate: 9.7% (N=250)
How can systems reason soundly with LLM knowledge in spite of its gluts and gaps?	**By using a sound and complete paraconsistent logic** with an LLM-grounded bilateral semantics	● F1: +0.062 with bilateral over unilateral factuality evaluation (N=800) ● Proof-of-concept tableau reasoner

Towards knowledge engineering as human-LLM dialogue



The stochastic parrots challenge assumes a *representationalist* stance: questioning if LLMs *internally understand the meaning of their output*

We argue instead for an *inferentialist* stance: that *whether LLMs understand their output doesn't matter;* what matters is whether they can play a meaningful role in a natural language dialogue with humans *in which meaning emerges from a game of giving and asking for reasons*

Our key findings provide evidence that LLMs *can* meaningfully play that role

This suggests that we can not just solve, but *dissolve*, the implementation problem

In conclusion



Caspar Barlaeus

“While others are counting their money and weighing it in a pair of scales, you should be weighing the words of wise men; ***while others are weighing bronze, pepper and flax, you should be weighing the importance of philosophy.”***

Barlaeus, C. (2025). Mercator Sapiens. In A. L. Post & C. Vermeulen (Eds.), *The Wise Merchant* (pp. 62–125). Routledge. (Original work published 1632)